

Outset · AI in Healthcare · Day 3 · Group 4

Good for whom?

Skin-cancer AI, access, and who is missing from the training data

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BACKGROUND

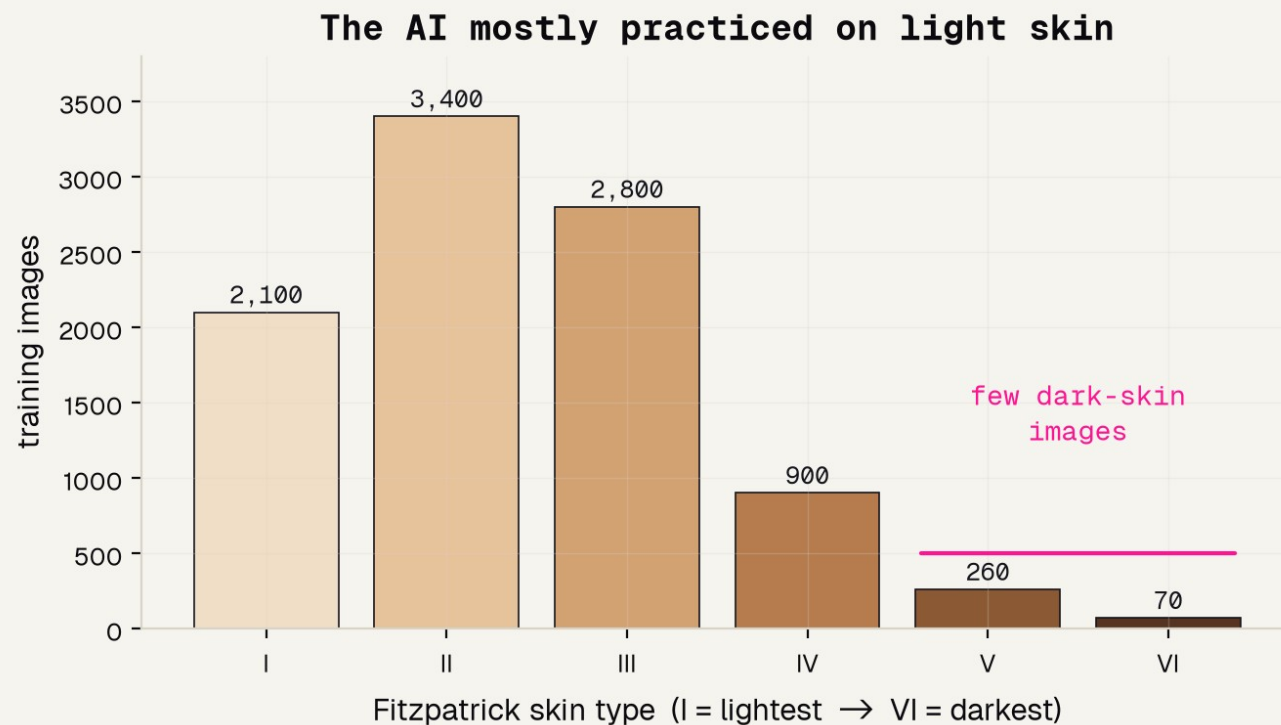
A promise, and a catch



A model only learns what it is shown.

Skin-cancer AI is pitched as a way to reach the roughly three billion people with little access to a dermatologist. That promise is real, and so is the catch: the images these models learn from are badly skewed toward light skin.

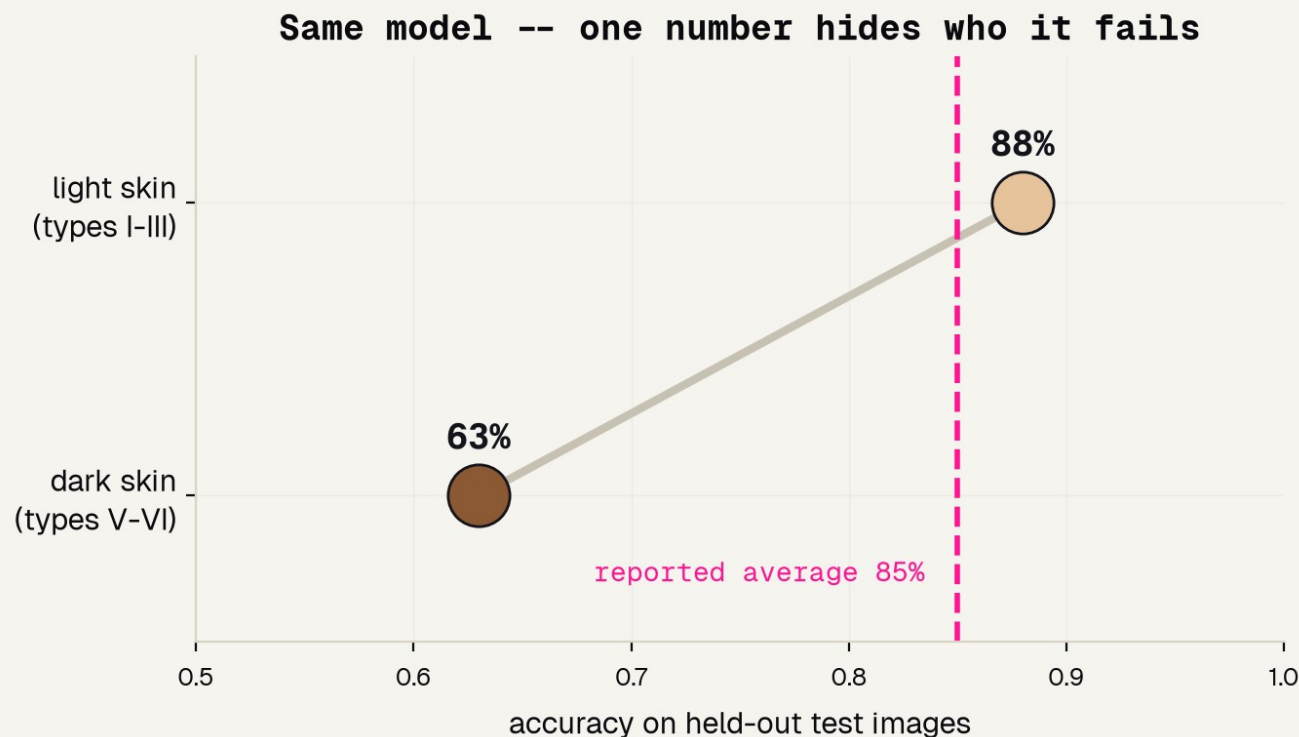
The AI mostly practiced on light skin



adapted from Wen et al. 2022 / Groh et al. 2021

- A review of twenty-one public skin datasets, over a hundred thousand images, found almost all came from Europe, North America, and Oceania, with almost none labeled as darker skin. On the Fitzpatrick scale the training data piles up on the left and nearly vanishes on the right.

Same model, one number hides who it fails

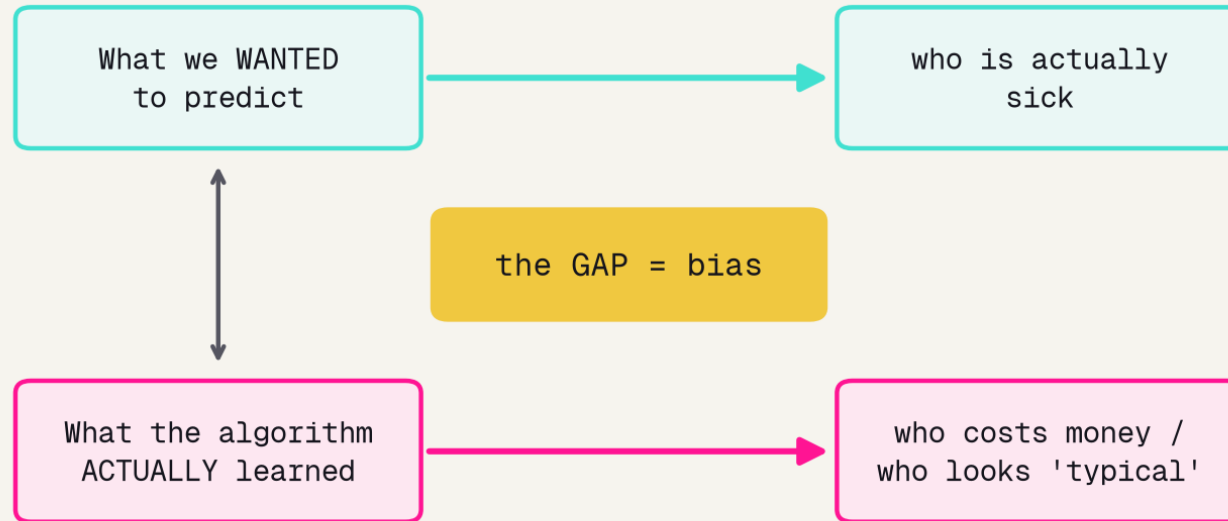


adapted from Daneshjou et al. 2022 / Groh et al. 2021

- Groh showed a model is most accurate on the skin types it saw most; Daneshjou showed top models drop sharply on dark skin. Same model, very different results depending on who you are, and the single average number a report shows sits in between and hides it.

Accurate on average can still be unfair

A model can score great on the wrong target



concept from Obermeyer et al. 2019

- Obermeyer studied a health algorithm used on millions of patients that looked fair on paper but underserved Black patients, because it was trained to predict cost as a stand-in for illness. It scored well on the wrong target. That is the exact suspicion we bring to our own model.

METHODS

The same model, a harder question



Change the question, not the model.

We reuse the screening group's exact DermaMNIST model -- a ResNet18 pretrained on ordinary photos, with a fresh head, sorting images into seven skin conditions. Same tool, harder question: not how good is it, but good for whom?

The equity lens: break the average apart



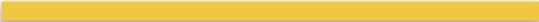
Good for whom?

A single accuracy number is a blend, and a blend can hide that one group is being carried by another. The screening group asks how good the model is; we refuse to trust the average and split the score apart by condition.

RESULTS

The headline number looks fine

68%

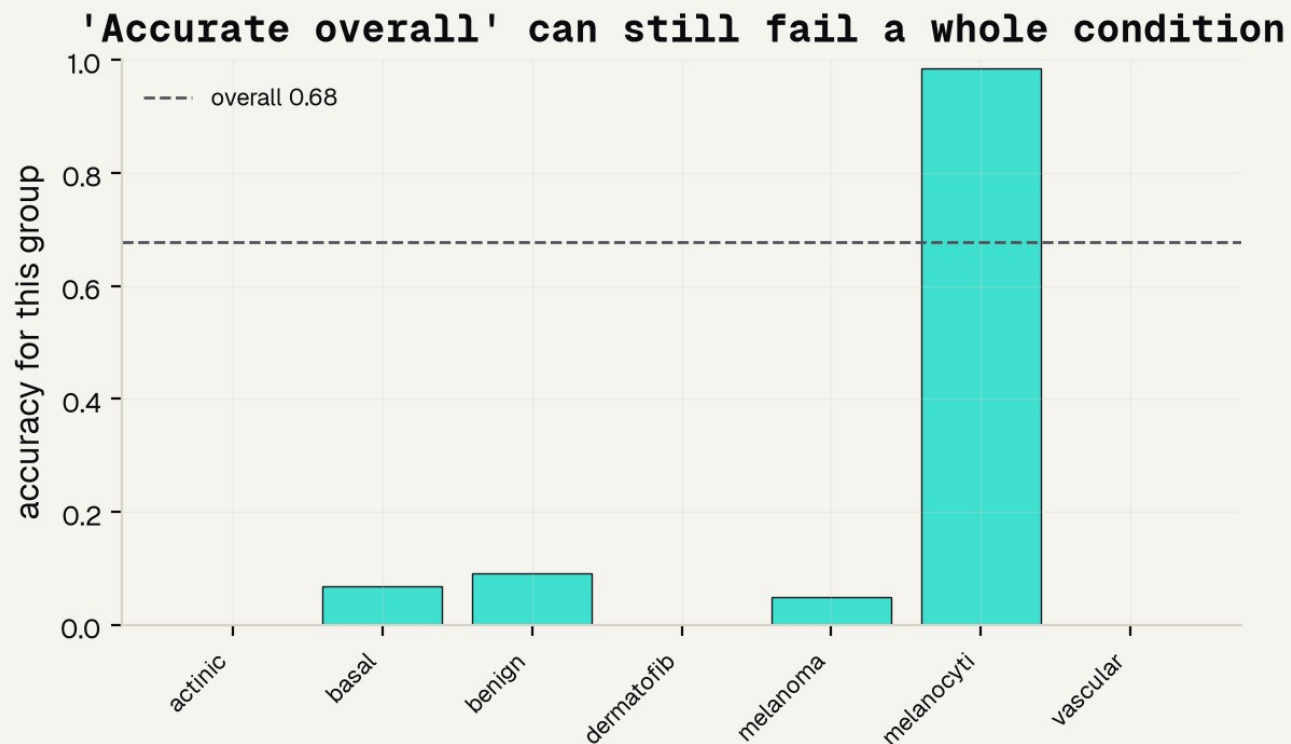


overall accuracy

across all seven conditions, blended into one number

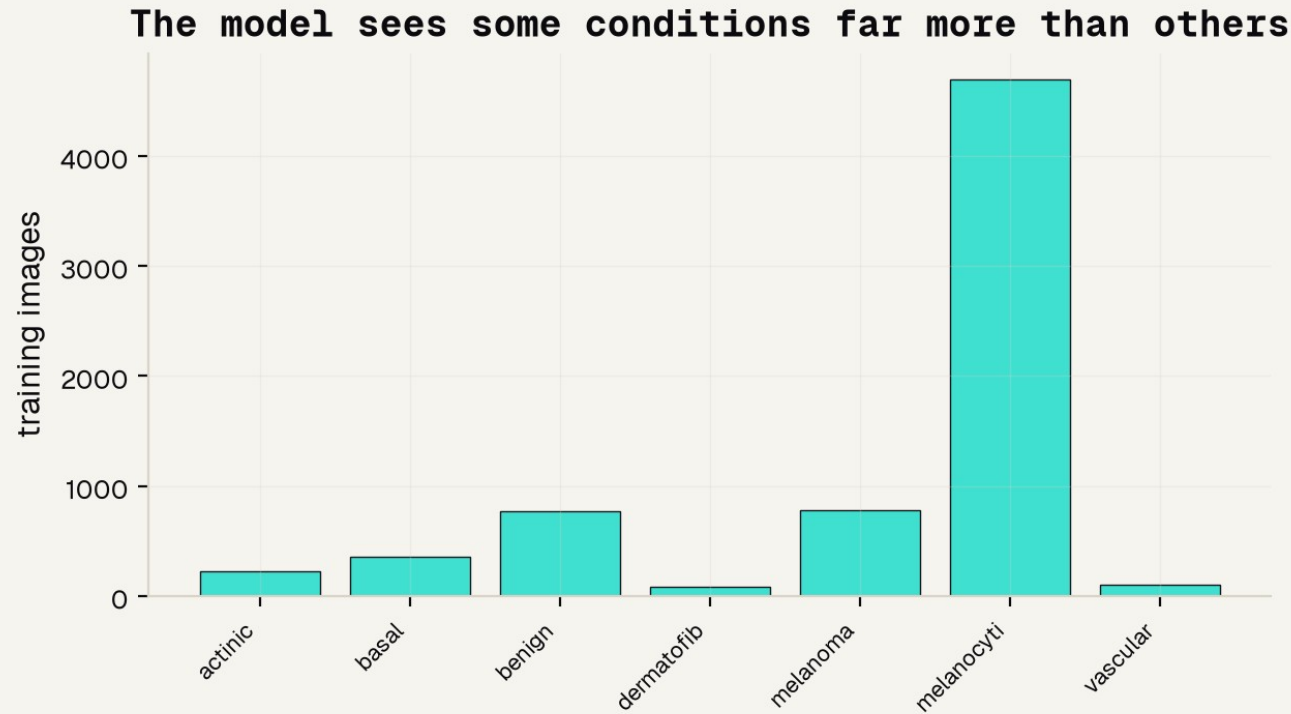
Not amazing, not embarrassing -- the kind of number you would quietly accept and move on. This is exactly the moment the equity lens says: stop, and break it apart.

Break it apart: a whole condition is failed



- Splitting the same score by condition, the average was hiding a near-total failure
- Ordinary moles, the giant class, are caught almost perfectly at ninety-nine percent and prop the average up. Several rarer conditions are caught almost never, some flat at zero.

Why: the model sees some conditions far more



- The gap is not mysterious
- One class has thousands of training images and the rare conditions have only dozens. Whatever the model sees most, it gets best at. This is the same skew the literature describes for skin tone, along the axis DermaMNIST actually labels.

The blind spot we cannot measure

Who is in the training data?

DermaMNIST records the lesion type -- but not the patient's skin tone, age, sex, or country. Dermatology datasets skew heavily toward lighter skin.

So we literally cannot check:

- ✗ Does it work as well on darker skin?
- ✗ Who was it never trained to see?
- ✗ Who gets helped -- and who gets missed?

- The gap the literature warns about most is along skin tone, and DermaMNIST ships no skin-tone labels. It records the lesion type and nothing about the patient. So we cannot check the real skin-tone gap here. We can demonstrate the risk of a blind spot, not measure it, and saying so is part of doing this honestly.

BEING HONEST

Who gets left out



Worst where care is scarcest.

The people the pitch was meant to help most are the ones most likely missing from the data -- disproportionately in regions and skin types barely present in training sets. A tool that is worst where care is scarcest can widen the gap it was sold to close.

References

1 Fitzpatrick 1988

The six-point skin-type scale (I-VI) dermatology AI now uses to describe who is in a dataset.

2 Groh et al. 2021

Fitzpatrick 17k: a model is most accurate on the skin types it saw most during training.

3 Daneshjou et al. 2022

Diverse Dermatology Images: state-of-the-art models drop sharply on dark skin tones.

4 Obermeyer et al. 2019

A health algorithm underserved Black patients by predicting cost instead of illness.

5 Wen et al. 2022

Review of 21 public skin datasets (106,950 images): almost all are light-skinned.

6 Adamson & Smith 2018

AI trained on fair-skinned archives can widen, not close, dermatology care gaps.

7 Buster et al. 2012

Skin-disease outcomes are worse for underserved, low-access, minority patients.

Thank You

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